School

of

Electronics and Communication Engineering

Internship Training Report

on

DASy (Driver Assistance System Controller)

By:
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SCHOOL OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to Certify that "Prajwal T P"(01FE20BEC218) studying in final year has undergone Industrial Internship Training in Bosch Global Software Technologies at Bengaluru from 2 February 2024 to 31 May 2024 in partial fulfillment for the award for Bachelor of Engineering in Electronics and Communication in the School of Electronics and Communication Engineering of KLE Technological University, Hubballi for the academic year 2023-2024.

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-Prajwal T P

ABSTRACT

Global industrial development has included the automobile industry as a significant sector. As technology has advanced, automobiles have been outfitted with an array of gadgets that enable them to carry out diverse tasks, enhance their degree of independence, and bestow upon their drivers a heightened sense of assurance and comfort . The advent of automated driving marks a transformative era in the automotive industry, promising unprecedented levels of safety, efficiency, and convenience on the road. However, as the complexity of automated driving functions continues to escalate, the limitations of current Electronic Control Units (ECUs) become increasingly evident. Recognizing this pressing need for innovation, Bosch has pioneered the development of the Driver Assistance System Domain Controller, known as DASy. DASy represents a significant shift in vehicle computing, offering an advanced solution to the growing demands of automated driving. Unlike traditional ECUs, which may struggle to handle the intricate algorithms and data processing requirements of advanced driving functions, DASy is specifically designed to excel in this area. By utilizing cutting-edge hardware and software technologies, DASy enables vehicles to navigate complex traffic scenarios with unmatched precision and reliability. At the core of DASy lies Bosch's unwavering commitment to pushing the boundaries of automotive innovation. Drawing upon decades of expertise in automotive electronics and systems integration, Bosch has developed DASy to serve as the foundation of next-generation automated driving platforms. Whether it's real-time sensor fusion, predictive analytics, or adaptive decision-making, DASy seamlessly manages a multitude of functions to ensure a smooth and safe driving experience. Additionally, DASy features a modular and scalable architecture, allowing automakers to adapt and evolve their automated driving systems with ease. As the automotive landscape continues to progress, DASy is poised to meet the changing needs of both manufacturers and consumers, ushering in a new era of mobility defined by intelligence, efficiency, and safety. In conclusion, DASy represents a pivotal advancement in the pursuit of fully automated driving. By addressing the shortcomings of conventional ECUs and providing a robust, future-proof computing platform, Bosch has established a new benchmark for vehicle computers. With DASy leading the way, the vision of a safer, smarter, and more sustainable future on the road is closer than ever before..

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Chapter 1

Introduction

This chapter concisely introduces about the company, objectives and scope of the internship.

1.1 About the Organization

As a trailblazer in technology and innovation, Bosch Global Software Technologies has shaped the software solution landscape across numerous sectors. Bosch Global Software Technologies was founded with the goal of transforming how companies function and communicate in the digital era. Since then, it has continuously provided cutting-edge software solutions and services that are customized to satisfy the changing demands of its wide range of clients.

Operating at the nexus of technology and industry, Bosch Global Software Technologies leverages its deep domain expertise to drive digital transformation across various sectors. As a subsidiary of the globally renowned Bosch Group, the company inherits a legacy of excellence and a commitment to quality that spans over a century.

Figure 1.1: Bosch Global Software Technologies

The core values of Bosch Global Software Technologies are the unwavering pursuit of innovation and the delivery of value-driven solutions that help businesses thrive in dynamic environments. Utilizing cutting-edge technical advances like artificial intelligence, machine learning, and the Internet of Things (IoT), the company is committed to developing software solutions that assist businesses in achieving operational excellence and seizing new possibilities.

The company operates in four primary business sectors, which are :

1. Solutions (both software and hardware)
2. Consumer products, such as power tools and home appliances
3. Industrial technology, encompassing control and driving
4. Building technology and energy

Figure 1.2: Business Model of Bosch

1.1.1 Bosch India

In the domains of consumer goods, energy and building technology, industrial technology, mobility solutions, and consumer goods, Bosch is a prominent provider of technology and services in India. Furthermore, Bosch maintains the largest non-German development center for end-to-end engineering and technology solutions in India. Since its establishment in 1953, Bosch has expanded its manufacturing operations in India to include 14 manufacturing facilities and 7 development and application centers.Globally, there is a growing need for Bosch.

In India, the Bosch Group employs about 29,000 people, and in 2014, it brought in a total of roughly Rs. 15,250 crores, of which Rs. 10,800 crores came from outside sources. The Group filed for about 150 patents in 2014, and it employs about 12,000 research and development colleagues in India. The automotive sector accounts for 84 percent of Bosch India's income, with the remaining 16 percent coming from its non-automotive businesses, which include consumer retail, power tools, energy and building solutions, packaging, and packaging. Bosch Ltd., Bosch Chassis Systems India Ltd., Bosch Rexroth India Ltd., Robert Bosch Engineering and Business Solutions Pvt. Ltd., Bosch Automotive Electronics India Pvt. Ltd., and Bosch are the nine companies that the Bosch Group works through in India.

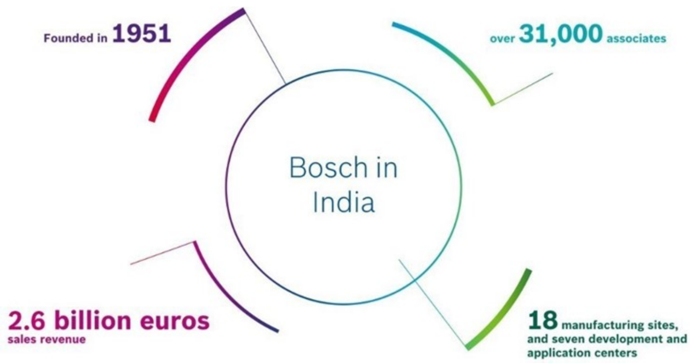

Figure 1.3: Bosch in India

1.1.2 Activities of Bosch

Approximately 60 percent of Bosch's yearly global sales are generated by automotive technology.The spark needed to ignite the fuel in the majority of the earliest internal combustion engines was produced by Bosch's invention of the first usable magneto, an early ignition electrical source that is still in use in general aviation engines. Bosch's corporate emblem until this moment displays the armature from a magneto. Bosch was an early manufacturer of Anti-lock Braking System (ABS), and as time passed, Bosch became a leader in such specialized fields as traction control systems (TCS), the Electronic Stability Program (ESP), body electronics (such as central locking, doors, windows and seats), and oxygen sensors, injectors and fuel pumps. Even in lowly technological domains like engine cooling fans, wiper blades, spark plugs, and other aftermarket Bosch sells more than 1 billion dollars worth of parts annually. In-vehicle navigation systems and car audio systems are two areas in which Bosch excels. Among the OEMs to receive hybrid diesel-electric technology from Bosch is PSA Peugeot 3008.

1.1.3 Industrial Technology

The subsidiary of Bosch As a provider of industrial technology, Bosch Rexroth makes electric, pneumatic, and hydraulic machinery for moving, driving, and controlling machines in a variety of industries, including mining and the automotive industry. For producers of food, candy, pharmaceutical, and related products, Bosch Packaging Technologies develops, plans, builds, and installs packaging lines. One of the biggest providers of packaging technologies is Bosch.

1.1.4 Consumer Goods and Power Tools

Bosch serves the consumer products and building technology sectors through its power tool, security system, and thermal technology divisions. It also operates a household appliance business under a joint venture between BSH Bosch and Siemens Hausger¨ate GmbH. Power tools are supplied in the United States by the Mt. Prospect, Illinois-based Robert Bosch Tool Corporation. Among the biggest producers of portable power tools globally, Bosch is known for its brands, which include Hawera, Skil, Dremel, RotoZip, Freud, Vermont American, and many more.

1.1.5 Security Systems

To expand their presence in the North American market for the production and distribution of security and life safety equipment, Bosch acquired Detection Systems and Radionics, Inc. in 2001. Bosch also acquired additional sales channels in Europe, Asia-Pacific (including Australia), and Latin America through the acquisition of Detection Systems. By purchasing Philips Communications and Security, Inc. in 2002, Bosch expanded its line of business to include sales channels and a video surveillance portfolio. The Telex Group, which included ElectroVoice, Dynacord, Midas, Klark Teknik, Telex, and RTS, was purchased by Bosch in January 2006. On December 8, 2009, Midas and Klark Teknik left Bosch and are currently a member of Music Group. To increase the range of products they offered for video surveillance, Bosch purchased the rugged camera and IP camera firm Extreme CCTV in 2008. October 2016, Bosch Security Systems, Inc. announced that the C-CURE 9000 security and event management platform from Software House from Tyco Security Products seamlessly integrates with its IP and high definition (HD) cameras and recording solutions.

1.1.6 Bosch Global Software Technologies

Robert Bosch GmbH owns all of the shares of Robert Bosch Engineering and Business Solutions Private Limited (RBEI). With more than 17,000 employees, RBEI is Bosch's largest software development center outside of Germany, proving that it is the company's technological leader in India. RBEI holds an ISO 9001:2008 certification (2012), has been assessed at CMMI-L5 according to version 1.3 (2011), is certified at ISO 27001, and is based on version 2.5 of ISO 15504-5 and 7. Its seven state-of-the-art facilities are located in Bengaluru and Coimbatore, India. RBEI mainly offers solutions in three areas to businesses:

1. Engineering Services
2. IT services
3. Business services

1.1.7 Vision, Mission and Goals

"Invented for connected life" is the BGSW's stated mission statement. The goal is to create and commercialize cutting-edge connected gadgets, customized Internet of Things (IoT) solutions, and automotive powertrains. to strengthen our software, sensor, and electronics capabilities in order to support new business models for international markets. to raise productivity, comfort, security, and quality of life in general. The company's objectives consist of:

- Focus on the Future and Outcomes.
- Accountability and Durability.
- Self-motivation and initiative.
- Trust and Transparency.

- Equitableness.
- Legality, Credibility, and Dependability.
- Variety.

The company invests significant resources in research and development (RD) activities to develop new products, technologies, and solutions that address emerging market needs and challenges. These activities showcase the diverse range of industries and technologies in which Bosch is actively involved. The company's focus on innovation, sustainability, and quality has contributed to its global reputation and position as a leading provider of technology and services. Bosch places a strong emphasis on RD to drive innovation and stay at the forefront of technological advancements.

1.2 Objectives

- Learning about the organization and understanding the structure of the organization.
- Knowing the teammates and working together to understand the teamwork and carrying out the task assigned as a team.
- Learning and understanding the training sessions planned to increase the technical knowledge and also the better understanding of tools used to carry out the projects.

1.3 Organization of the report

Chapter 2 includes the background study required for vehicle communication, about CAN protocol

Chapter 3 deals with ECU Programming about DASy (IDC-5 Project) used to carry-out my task

Chapter 4 deals with the tools learnt during the internship

Chapter 5 includes the learning outcomes and developed skills during the training

Chapter 6 includes some of the challenges faced and the solution drawn for the problem

Chapter 7 draws conclusions of the internship traing carried out at the company.

Chapter 2

CAN - Controller Area Network

Robert Bosch created the Controller Area Network (CAN) protocol sometime around 1986. A standard called the CAN protocol was created to guarantee that a microcontroller and other devices could communicate with one another without the assistance of a host computer. The broadcast kind of bus is one of the distinguishing characteristics of the CAN protocol over other communication protocols. Broadcasting is the process of sending the gathered data to every node. A node in this context could be an Ethernet port or USB cable-equipped sensor, gateway, or microcontroller that enables network communication between the computer and other devices. Since the CAN is essentially a message-based protocol, priority is determined by looking up a message identification that is included with each message. Within the CAN network, node identity is not required, making it extremely simple to add and remove nodes from the network. This kind of serial half-duplex communication technique is asynchronous. The CAN network is a two-wired communication protocol that is linked by a two-wired bus. Although it was initially intended for use in cars for communication, it is currently utilized in a variety of other settings, such as KWP 2000 and UDS. Additionally, on-board diagnostics utilize CAN.

2.1 Need for CAN

The requirement for a centralized common communication protocol grew along with the number of electronic devices. For example, a modern car may include more than seven TCUs for various subsystems, such as the dashboard, transmission control, engine control unit, and many more. tremendous-speed communication would be achieved if all of the nodes were connected one to one, but this would come at a tremendous cost in terms of complexity and wiring. The CAN LOW and CAN HIGH wires are the only two required for CAN, which was introduced as a centralized solution. The message prioritizing and flexibility of the CAN protocol, which allows a node to be added or deleted without negatively impacting the network, contribute to its extremely high efficiency.

2.2 Applications of CAN protocol

Because of the functionality it provides, the CAN protocol is now utilized in many other fields, although its initial purpose was to address the communication problem that arises within automobiles. Some of these other fields that have adopted the CAN protocol include:

1. Automotive, encompassing buses, trucks, and passenger cars etc.
2. Aviation and navigational electronic equipment
3. Industrial automation and mechanical control
4. Elevator and escalator; Building automation
5. Medical instruments and equipment
6. Marine, medical, industrial, and medical

2.3 CAN Frame - Standard and Extended Data Frame Format

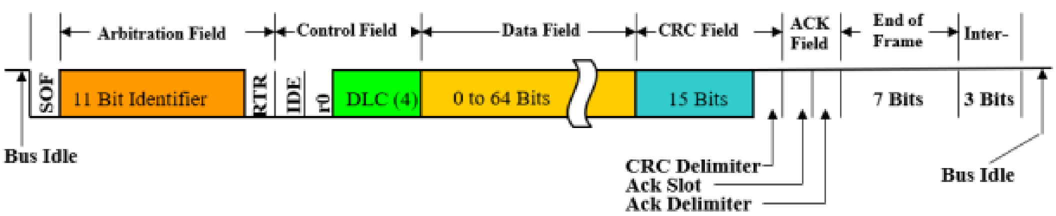

Figure 2.1: Standard Data Frame Format

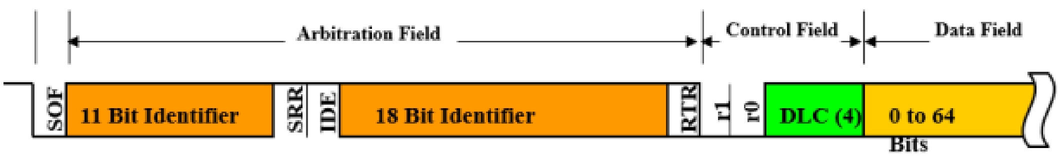

Figure 2.2: Extended Data Frame Format

Start of Frame, or SOF: A network's new frame has been entered, according to SOF. It has just one bit in it.

Identifier The 11-bit or 29-bit (11+18 bit) identifier is a field that is used to uniquely identify the message that is being transmitted. In CAN networks, the identification establishes the message priority and makes message filtering easier. Standard identification refers to the 11-bit identification, and Extended Identifier refers to the 29-bit identifier.

Control Symbols: Various control bits that provide further details about the frame are contained in the Control field. The Identifier Extension (IDE) bit, which differentiates between Standard and Extended IDs, and the Remote Transmission Request (RTR) bit, which specifies whether the frame is a data frame or a remote frame, are two examples of the control bits.

Code for Data Length (DLC): The number of bytes of data being transmitted in the frame is specified in the DLC field. It can indicate the length of the data payload with a value between 0 and 8.

Data Field The actual data being transferred is contained in the Data Field. The DLC value determines the size of the data field, which can be anywhere between 0 and 8 bytes. CRC Field: Errors in the transmitted data are found using the CRC (Cyclic Redundancy Check) field. Depending on the frame type and identifier length, it is a 15- or 17-bit field.

Acknowledgment An Acknowledgment (ACK) bit is sent by the receiving node to confirm correct reception of a Data Frame once it has been successfully received. A transmitting node retransmits the frame if it does not receive an ACK in a predetermined amount of time.

Bit Timing: The highest priority node to transmit is determined by CAN using a bit-wise arbitration method. The bit rate, which controls the length of each bit and the synchronization between network nodes, defines the bit timing.

Error Detection and Handling: Error handling and detection techniques for bit, frame, and stuff problems are part of CAN. Error frames are produced when errors happen to let other nodes know about the error condition.

EOF (End of Frame): EOF consists of 7 consecutive recessive bits.

The message is dispatched in accordance with the arbitration field's priority setting. In the case of an extended frame, the message identification is 29 bits, whereas it is 11 bits for a conventional frame. As a result, the message identification can be created by the system designer at the design level. This would also imply that a smaller message identification would accompany a higher message priority and vice versa. To send the message, the sender must wait for the CAN bus to become idle. The sender transmits the dominating bit, or the SOF, for bus access if the CAN bus is discovered to be idle. The most important bit is then sent together with the message identifying bit. Should the node identify the prominent area of the bus The node ceases transmitting additional bits after transmitting the recessive bit, indicating that it has lost the arbitration. When the bus is free, the sender will wait and try again with the message.

Chapter 3

DASy - Driver Assistance System Controller

In the rapidly advancing field of automotive technology, the exciting idea of automated driving—which offers a vision of safer, more efficient transportation—beckons. However, a major barrier emerges as this future gets closer: the current generation.

Because driving activities are becoming more complicated, Electronic Control Units (ECUs) may become unstable. Leading innovator in automotive engineering, Bosch, sees the need for change and delivers a revolutionary solution in the form of the Driver Assistance System Domain Controller, or DASy. make a mental image of the route ahead and decide on the best driving strategies in real time. Once in DASy's custody, this preprocessed data forms the basis of an advanced accuracy 360-degree depiction of the surroundings around the car. DASy incorporates a comprehensive view of the environment through methodical processing and analysis, giving the car unparalleled situational awareness. DASy's integrated functionalities use this model as a canvas to paint a vision of the future and identify the optimal driving strategies in real time. Not only does DASy have a high bandwidth capacity, but it also includes a strong processor powered by a cutting-edge microprocessor. With processing capability greater than 3000 DMIPS, DASy can easily manage the complexities of daily driving conditions. It accomplishes this by coordinating an intricate web of algorithms to produce a thorough picture of the path ahead. A brilliant example of how technology is progressing toward completely autonomous driving is DASy. Unlike its predecessors, which often struggle to manage the complex algorithms and data streams included in modern driving systems, DASy is built expressly for this task. Modern hardware and software are used by DASy to handle the complexity of today's traffic circumstances with unparalleled accuracy and efficiency.

3.1 Hardware Variants

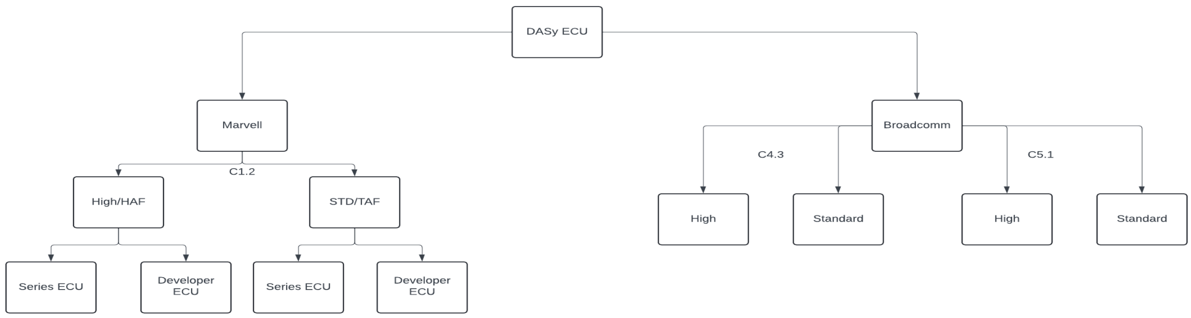

Figure 3.1: Hardware Variants hierarchy

Marvell and Broadcom are the two main versions of the DASy ECU. The hardware components are where they diverge most. Most notably, they are distinguished by the oscillator switch. These variations meet various needs and use cases in the automobile industry. **Marvell Subtypes:** High (HAF): This Marvell ECU, sometimes referred to as the high-performance edition, offers improved features. Standard (TAF): Without specific specifications, the standard Marvell ECU is appropriate for most applications. Marvell ECUs are frequently used by developers to optimize and fine-tune their applications. **Variants of Broadcom:** The Broadcom versions are made especially to fulfill certain requirements in the architecture of the car. Though not stated clearly, these variations probably provide special features or optimizations.

Developer ECU (Electronic Control Unit):

- Purpose: The developer ECU, also known as a test or prototype ECU, is primarily used during the development and testing phases of automotive electronic systems.
- Testing and Validation: Developers use this ECU to validate software, perform integration tests, and verify system behavior. Debugging and Diagnostics: It allows engineers to debug code, monitor sensor data, and diagnose issues.
- Simulation: Developers can simulate various scenarios (e.g., engine load, temperature changes) to assess system responses.
- Hardware and Software: Typically, developer ECUs have additional debugging interfaces (e.g., JTAG, BDM) for direct access to the microcontroller. They may run specialized software tools for debugging and testing.
- Not for Production: Developer ECUs are not suitable for production vehicles due to their additional features and interfaces.
- Not for Production: Developer ECUs are not suitable for production vehicles due to their additional features and interfaces.

Series ECU (Production ECU):

- Purpose: Series ECUs are the production-ready electronic control units installed in vehicles.
- Control and Regulation: Series ECUs manage various functions (e.g., engine control, transmission, ABS, airbags).
- Efficiency and Safety: They optimize performance, fuel efficiency, and safety.
- Real-World Operation: Series ECUs operate under real-world conditions.
- Hardware and Software: Series ECUs are optimized for cost, size, and reliability. They use production-grade microcontrollers and standard communication protocols (e.g., CAN, LIN).
- Integration: Series ECUs are integrated into the vehicle's network and communicate with other ECUs.
- Calibration: Manufacturers calibrate series ECUs to meet specific vehicle requirements.

Chapter 4

Tools Used

4.1 CANoe

CANoe (Controller Area Network Open Environment) is a highly advanced software tool developed by Vector Informatik, designed to cater to the diverse needs of automotive engineers throughout the development lifecycle of electronic control units (ECUs) and in-vehicle networks. As a multifaceted tool, CANoe provides a comprehensive suite of functionalities for simulation, analysis, testing, diagnostics, and maintenance, making it an essential asset in the automotive industry.

One of CANoe's primary strengths lies in its ability to simulate and test entire vehicle networks, including individual ECUs. It supports a wide range of communication protocols, such as CAN (Controller Area Network), LIN (Local Interconnect Network), FlexRay, Ethernet, and MOST (Media Oriented Systems Transport). This broad protocol support ensures that CANoe can be used across various types of automotive networks, making it versatile and adaptable to different project requirements.

In terms of development and analysis, CANoe offers powerful tools for designing and analyzing network communications. Engineers can monitor and analyze data traffic, examine message contents, and assess timing characteristics in real-time. This capability is crucial for understanding how different components within a vehicle communicate and interact, ensuring that the network operates efficiently and effectively. The tool's graphical and text-based analysis options allow for detailed examination and troubleshooting of communication issues.

CANoe's diagnostic capabilities are another significant aspect of its functionality. It provides robust tools to troubleshoot and verify the correct functioning of ECUs and entire networks. For instance, it supports diagnostic protocols like Unified Diagnostic Services (UDS), which are essential for identifying and resolving issues within the vehicle's electronic systems. These diagnostic features enable engineers to perform comprehensive tests and diagnostics, ensuring that the ECUs and networks meet all required standards and specifications.

Extensibility and customization are key features that set CANoe apart. Users can extend its capabilities by writing custom scripts and programs using the CAPL (Communication Access Programming Language). This allows for the automation of repetitive tasks, customization of testing procedures, and the creation of complex simulations tailored to specific project needs. Additionally, CANoe can be integrated with other software tools through APIs, facilitating seamless workflows and enhancing its functionality within the broader toolchain used in automotive development.

CANoe is utilized across various stages of vehicle development, from initial design and prototyping to production and field diagnostics. During the design phase, engineers use CANoe to model and simulate network behavior, ensuring that the proposed design meets performance criteria. In the prototyping phase, CANoe helps in refining and validating the design through extensive testing and analysis. Once the vehicle is in production, CANoe continues to be a valuable tool for quality assurance and compliance testing. Even after vehicles are on the road, CANoe aids in field diagnostics, helping service technicians diagnose and repair network-related issues efficiently.

In summary, CANoe is a vital tool in the automotive industry, providing a unified environment for handling the complexities of vehicle networks. Its comprehensive features for simulation, analysis, testing, diagnostics, and extensibility make it indispensable for ensuring the reliability, efficiency, and safety of automotive electronic systems. By enabling engineers to thoroughly test and validate network communications and ECUs, CANoe plays a crucial role in the development of modern, sophisticated vehicles.

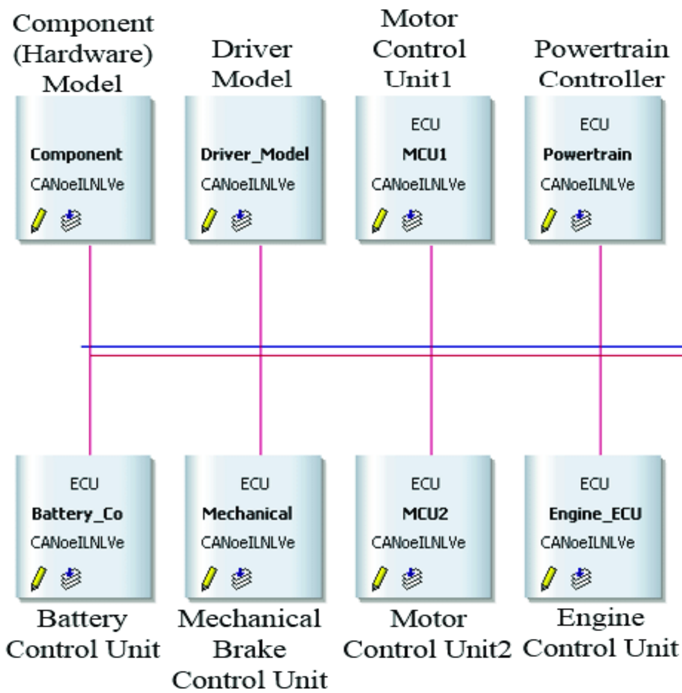

Figure 4.1: Hardware Variants hierarchy

4.2 RQM

Rational Quality Manager (RQM) is a comprehensive, web-based test management tool developed by IBM, designed to streamline the planning, management, and execution of quality assurance activities in software development. It facilitates the creation of detailed test plans that outline testing strategies, scopes, objectives, resources, schedules, and environments, ensuring a structured and organized approach to testing. With robust test case management capabilities, RQM allows users to define, organize, and manage test cases, including specifying test steps and expected results, and linking test cases to requirements for comprehensive coverage and traceability. The tool supports both manual and automated testing, integrating with various test automation tools and frameworks to facilitate test execution and scheduling. RQM's integration with defect tracking systems allows for efficient logging and management of defects discovered during testing, ensuring prompt resolution by tracking their lifecycle from discovery to closure. Additionally, RQM offers powerful reporting and analytics features, providing insights into test progress, coverage, and quality metrics through customizable dashboards and reports, enabling stakeholders to make informed decisions based on real-time data. The tool enhances collaboration among team members by providing a centralized platform for sharing test artifacts and communicating, and it integrates seamlessly with other tools in the IBM Engineering Lifecycle Management (ELM) suite as well as third-party tools, making it highly versatile and beneficial in diverse development environments. Overall, RQM is designed to enhance the efficiency and effectiveness of the test management process, ensuring that software products meet the highest quality standards through rigorous quality management practices.

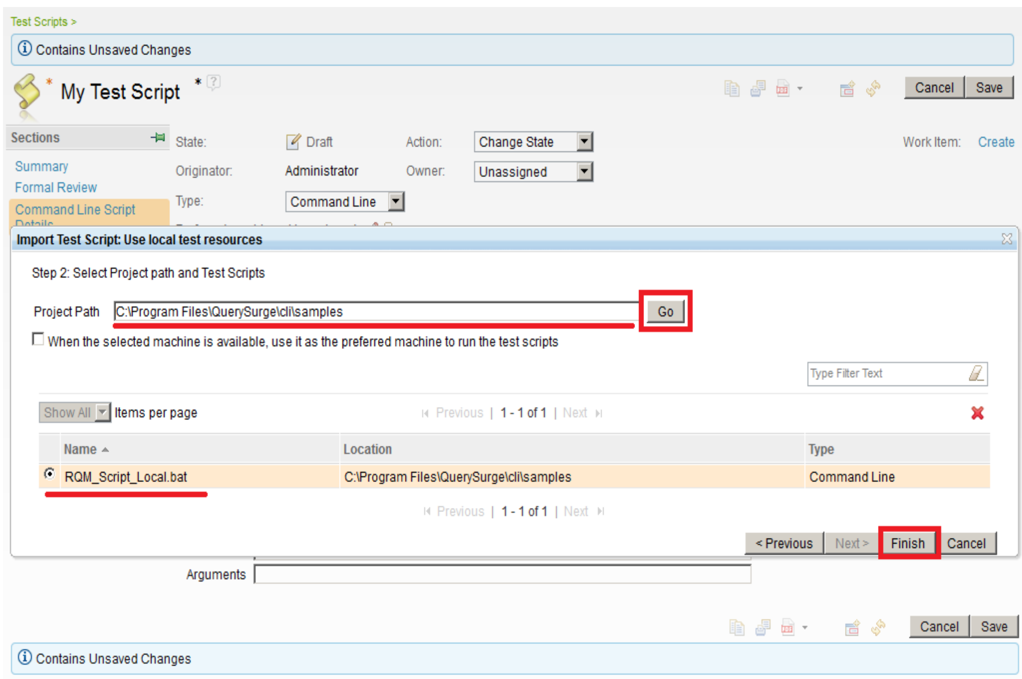

Figure 4.2: Hardware Variants hierarchy

4.3 Doors Next Generation

DOORS Next Generation (DNG) is a sophisticated requirements management tool developed by IBM, designed to streamline the process of capturing, organizing, and managing requirements throughout the development lifecycle, ensuring alignment with business goals and compliance with industry standards. It allows users to create and manage requirements in various formats, such as text, tables, and diagrams, and to categorize and prioritize them effectively. DNG promotes collaboration by providing a platform where stakeholders can review and contribute to requirements in real time, enhancing communication and reducing misunderstandings. The tool ensures full traceability of requirements by linking them to other project artifacts like design documents, test cases, and change requests, which facilitates impact analysis and ensures comprehensive coverage and testing of requirements. Additionally, DNG supports version control and configuration management, enabling teams to track changes, compare different versions, and revert to previous states if necessary, thus maintaining a detailed historical record of requirements evolution. It is highly customizable to fit specific project needs and integrates seamlessly with other tools in the IBM Engineering Lifecycle Management (ELM) suite and third-party applications, promoting a cohesive workflow across various development phases. DNG also offers robust reporting and analytics capabilities, providing insights into the status and progress of requirements through customizable dashboards and reports, aiding stakeholders in making informed, data-driven decisions. Furthermore, the tool supports compliance with industry standards and regulations, ensuring that projects meet required specifications and legal obligations. Overall, DOORS Next Generation enhances the efficiency and effectiveness of requirements management, improves team collaboration, and ensures that projects are delivered successfully while adhering to all specified requirements and standards.

Chapter 5

Learning Outcomes

5.1 Skills and Knowledge Gained during Internship

1. During my internship I gained valuable skills, especially in understanding the ADAS and Autonomous driving field, Hardware interconnections of ECU (Test Bench),Different testing modules used in ECU testing.

2. I was able to relate my theoretical knowledge gained in the academics is used in real world projects.

3. The background knowledge required to carry out the data log book test and SAR test

4. I was able to learn in depth about the ECU programming modules such as Robustness flash loop test, Extended Flash loop test.

5. Reading and analysing requirements and documents in DNG.

5.2 Personal and professional development

1. **Technical Proficiency:** significantly enhanced my technical skills by gaining in-depth knowledge of vehicle communication protocols like CAN and diagnostics protocol like UDS.

2. **Communication Skills:** Engaging and communicating with team members and interacting with professionals in the automotive industry has improved my communication skills.

3. **Adaptability and Flexibility:** Working in a dynamic environment exposed me to various tasks and situations, fostering adaptability and flexibility. I learnt to adjust to new tools, technologies, and project requirements.

4. **Time Management:** Balancing multiple tasks during the internship honed your time management skills

5. **Professional Networking:** Building relationships with industry professionals and mentors provided valuable insights and guidance for my career growth.

Chapter 6

Challenges Faced

6.1 Difficulties encountered during the internship

1. **Tools installation:** Not all tools have clear, beginner-friendly documentation. Understanding how to configure, use, and troubleshoot the tool can be frustrating without proper guidance.It was time taking And it was difficult to understand how to request for services in IT service portal.

2. **Lack of Administrative Privileges:** If you're working on a company-provided machine, you might not have administrative privileges to install software. In such cases, we needed to request IT support or find workarounds.

3. **V Cycle Model:** Not adhering to the proper V-cycle during initial phase of task execution resulted in increased time consumption, dependencies on other modules, and a lack of flexibility in task execution.

6.2 How the challenges were addressed

1. As a novice, I had to ask my colleagues and college students who were already working on the tools how to install them and requested them to assist me.

2. It is necessary to seek for the rights to access certain things because, as external employees, we now lack permission to access them.

3. We were given a basic assignment to learn how to design, follow software architecture, code, and test as part of one of the manager's jobs. It was therefore exceedingly difficult to get the desired results when these stages were not followed in the correct order. Thus, adhering to the correct V Model or a particular design is crucial.

Chapter 7

Conclusion

7.1 Conclusion

Core competencies are critical in all industries. Without them, even simple things become difficult. Strong technical skills are therefore advantageous since they make complex systems easier to understand and help build the advanced skills required for career advancement in a particular profession.

7.2 Overall assessment of the internship experience

1. Through this internship I have enhanced my knowledge on ADAS, Autonomous driving , DASy ECU, Programming modules and Test modules.

2. I have learnt to use the tools and software that are required to carry out the project.

3. Communication with teammates, sharing the work, following a common design model all are the necessary actions during the task.

4. Self learning is important in the corporate world as it increases our knowledge and also utilization of time when any task is not assigned.

5. During this internship I was assigned with test cases execution and analysis to get familiar with CANoe tool.

6. In addition, I was put in a team to support complete the project's DNG documentation requirements.

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